BUS TICKET BOOKING

Aim

To develop an online bus ticket booking system that allows users to register, log in, search for buses, check seat availability, book and cancel tickets, post comments, and view booking history. It also allows administrators to manage buses, bus categories, users, schedules, and reports.

CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Bus Ticket Booking System</title>

  <style>
    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
      font-family: Arial, sans-serif;
    }

    body {
      background: #f3f6fa;
      color: #333;
    }

    header {
      background: #123c69;
      color: white;
      padding: 20px;
      text-align: center;
    }

    nav {
      background: #24527a;
```

```
  padding: 14px;
  text-align: center;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 8px;
  display: inline-block;
}

section {
  padding: 30px 6%;
}

h2 {
  text-align: center;
  color: #123c69;
  margin-bottom: 20px;
}

.container {
  max-width: 850px;
  margin: auto;
  background: white;
  padding: 22px;
  border-radius: 8px;
  box-shadow: 0 3px 10px #ddd;
}

input, select, textarea, button {
  width: 100%;
  padding: 11px;
  margin: 8px 0;
```

```
  border: 1px solid #ccc;
  border-radius: 5px;
}

button {
  background: #123c69;
  color: white;
  border: none;
  cursor: pointer;
}

button:hover {
  background: #2874a6;
}

.bus-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(230px, 1fr));
  gap: 18px;
}

.bus-card {
  background: white;
  padding: 20px;
  border-radius: 8px;
  box-shadow: 0 3px 10px #ddd;
}

.bus-card h3 {
  color: #2874a6;
  margin-bottom: 12px;
}

.bus-card p {
```

```
    margin: 8px 0;
}

.seat-layout {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 8px;
    margin: 15px 0;
}

.seat {
    padding: 12px 4px;
    background: #d9f2df;
    color: #155724;
    text-align: center;
    border-radius: 5px;
    cursor: pointer;
    border: 1px solid #8ac99b;
}

.seat.selected {
    background: #2874a6;
    color: white;
}

.seat.booked {
    background: #e57373;
    color: white;
    cursor: not-allowed;
}

.message {
    margin-top: 12px;
    padding: 12px;
```

```
    background: #e8f5e9;
    border-radius: 5px;
    display: none;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
    background: white;
}

th, td {
    border: 1px solid #ddd;
    padding: 10px;
    text-align: left;
}

th {
    background: #123c69;
    color: white;
}

.table-wrap {
    overflow-x: auto;
}

footer {
    background: #123c69;
    color: white;
    text-align: center;
    padding: 20px;
}
```

```
    @media (max-width: 600px) {
      section {
        padding: 20px 4%;
      }

      header h1 {
        font-size: 24px;
      }
    }
  </style>
</head>

<body>

<header>
  <h1>Online Bus Ticket Booking</h1>
  <p>Search, Select, Book and Travel</p>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#register">Register / Login</a>
  <a href="#search">Search Buses</a>
  <a href="#booking">Book Ticket</a>
  <a href="#history">My Bookings</a>
  <a href="#comments">Comments</a>
  <a href="#admin">Admin</a>
</nav>

<section id="home">
  <div class="container">
    <h2>Welcome to Bus Booking</h2>
    <p>Book your bus tickets online. Search routes, check available
      seats, select travel dates, and manage your bookings easily.</p>
```

```
  </div>
</section>

<section id="register">
  <div class="container">
    <h2>User Registration</h2>

    <form id="registerForm">
      <input id="regName" placeholder="Full Name" required>
      <input id="regEmail" type="email" placeholder="Email Address" required>
      <input id="regPassword" type="password"
          placeholder="Create Password" required>
      <button type="submit">Register</button>
    </form>

    <h2>User Login</h2>

    <form id="loginForm">
      <input id="loginEmail" type="email"
          placeholder="Email Address" required>
      <input id="loginPassword" type="password"
          placeholder="Password" required>
      <button type="submit">Login</button>
    </form>

    <button onclick="changePassword()">Change Password</button>
    <button onclick="forgotPassword()">Forgot Password</button>

    <p id="userMessage" class="message"></p>
  </div>
</section>

<section id="search">
  <h2>Search Buses</h2>
```

```
  <div class="container">
    <form id="searchForm">
      <label>From</label>
      <input id="from" placeholder="Departure city" required>

      <label>To</label>
      <input id="to" placeholder="Destination city" required>

      <label>Travel Date</label>
      <input id="travelDate" type="date" required>

      <button type="submit">Search Buses</button>
    </form>
  </div>
</section>

<section id="buses">
  <h2>Available Buses</h2>
  <div class="bus-list" id="busList"></div>
</section>

<section id="booking">
  <div class="container">
    <h2>Book Your Ticket</h2>

    <p>Select an available seat from the seat layout.</p>
    <div class="seat-layout" id="seatLayout"></div>

    <p id="selectedSeats">Selected Seats: None</p>

    <form id="bookingForm">
      <label>Passenger Name</label>
      <input id="passengerName" required>
```

```
        <label>Passenger Age</label>
        <input id="passengerAge" type="number" min="1" max="120" required>

        <label>Email</label>
        <input id="passengerEmail" type="email" required>

        <label>Bus</label>
        <select id="bookingBus" required></select>

        <button type="submit">Confirm Demo Booking</button>
      </form>

      <p id="bookingMessage" class="message"></p>
    </div>
  </section>

  <section id="history">
    <div class="container">
      <h2>My Booking History</h2>
      <div class="table-wrap">
        <table>
          <thead>
            <tr>
              <th>Booking ID</th>
              <th>Passenger</th>
              <th>Bus</th>
              <th>Travel Date</th>
              <th>Seats</th>
              <th>Fare</th>
              <th>Status</th>
              <th>Action</th>
            </tr>
          </thead>
```

```
          <tbody id="historyTable">
            <tr><td colspan="8">No bookings available.</td></tr>
          </tbody>
        </table>
      </div>
    </div>
</section>

<section id="comments">
  <div class="container">
    <h2>Comments and Reviews</h2>

    <form id="commentForm">
      <label>Select Bus</label>
      <select id="commentBus"></select>

      <label>Your Comment</label>
      <textarea id="commentText" rows="4"
          placeholder="Write your feedback..." required></textarea>

      <button type="submit">Post Comment</button>
    </form>

    <div id="commentList"></div>
  </div>
</section>

<section id="admin">
  <div class="container">
    <h2>Administrator Login</h2>

    <form id="adminLogin">
      <input id="adminPassword" type="password"
          placeholder="Admin Password" required>
```

```
  <button type="submit">Login as Admin</button>
</form>

<div id="adminPanel" style="display:none;">
  <h2>Manage Bus Information</h2>

  <form id="busForm">
    <label>Bus Name</label>
    <input id="busName" required>

    <label>Bus Category</label>
    <select id="busCategory">
      <option>Volvo AC</option>
      <option>Sleeper Coach</option>
      <option>Non-AC Coach</option>
      <option>Seater Coach</option>
    </select>

    <label>From</label>
    <input id="busFrom" required>

    <label>To</label>
    <input id="busTo" required>

    <label>Departure Time</label>
    <input id="busTime" type="time" required>

    <label>Fare per Seat (₹)</label>
    <input id="busFare" type="number" min="1" required>

    <label>Total Seats</label>
    <input id="busSeats" type="number" min="1" max="60" required>

    <button type="submit">Add New Bus</button>
```

```
        </form>

        <h2>Manage Existing Buses</h2>
        <div id="adminBusList"></div>

        <h2>Booking Reports</h2>
        <button onclick="generateReport()">Generate Booking Report</button>
        <div id="report"></div>

        <h2>Registered Users</h2>
        <div id="userList"></div>
      </div>
    </div>
</section>

<footer>
    <p>&copy; 2026 Online Bus Ticket Booking System</p>
    <p>Internet Programming Lab Experiment</p>
</footer>

<script>
    let users = [];
    let currentUser = null;
    let bookings = [];
    let comments = [];
    let nextBookingId = 1;
    let selectedSeats = [];
    let selectedBusId = null;

    let buses = [
      {
        id: 1,
        name: "Express Volvo",
        category: "Volvo AC",
```

```
    from: "Chennai",
    to: "Bangalore",
    time: "21:00",
    fare: 900,
    totalSeats: 20,
    bookedSeats: []
  },
  {
    id: 2,
    name: "Dream Sleeper",
    category: "Sleeper Coach",
    from: "Chennai",
    to: "Hyderabad",
    time: "20:00",
    fare: 1100,
    totalSeats: 20,
    bookedSeats: []
  },
  {
    id: 3,
    name: "City Travels",
    category: "Non-AC Coach",
    from: "Bangalore",
    to: "Chennai",
    time: "19:30",
    fare: 650,
    totalSeats: 20,
    bookedSeats: []
  }
];

function showMessage(id, message) {
  const element = document.getElementById(id);
  element.textContent = message;
```

```
    element.style.display = "block";
}

function escapeHTML(value) {
    return String(value).replace(/[&<>"']/g, character => ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        '"': "&quot;",
        "'": "&#39;"
    })[character]);
}

document.getElementById("registerForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const name = document.getElementById("regName").value.trim();
    const email = document.getElementById("regEmail").value.trim().toLowerCase();
    const password = document.getElementById("regPassword").value;

    if (users.some(user => user.email === email)) {
        showMessage("userMessage", "Email is already registered.");
        return;
    }

    users.push({name, email, password});
    showMessage("userMessage", "Registration successful. Please log in.");
    this.reset();
    renderUsers();
});

document.getElementById("loginForm").addEventListener("submit", function(e) {
    e.preventDefault();
```

```
  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u => u.email === email && u.password === password);

  if (!user) {
    showMessage("userMessage", "Invalid login details. Register first.");
    return;
  }

  currentUser = user;
  document.getElementById("passengerName").value = user.name;
  document.getElementById("passengerEmail").value = user.email;

  showMessage("userMessage", "Welcome, " + user.name + "!");
  renderHistory();
});

function changePassword() {
  if (!currentUser) {
    alert("Please log in first.");
    return;
  }

  const password = prompt("Enter your new password:");
  if (!password || password.length < 6) {
    alert("Password must contain at least 6 characters.");
    return;
  }

  currentUser.password = password;
  alert("Password changed in this demo.");
}
```

```
function forgotPassword() {
  alert("Demo only: password recovery must be implemented securely on a server.");
}

function renderBuses() {
  document.getElementById("busList").innerHTML = buses.map(bus => `
    <div class="bus-card">
      <h3>${escapeHTML(bus.name)}</h3>
      <p>Category: ${escapeHTML(bus.category)}</p>
      <p>Route: ${escapeHTML(bus.from)} to ${escapeHTML(bus.to)}</p>
      <p>Departure: ${escapeHTML(bus.time)}</p>
      <p>Fare: ₹${bus.fare}</p>
      <p>Available seats: ${bus.totalSeats - bus.bookedSeats.length}</p>
      <button onclick="selectBus(${bus.id})">Select Bus</button>
    </div>
  `).join("");

  updateBusOptions();
  renderAdminBuses();
}

function updateBusOptions() {
  const options = buses.map(bus =>
    `<option value="${bus.id}">${escapeHTML(bus.name)} - ${escapeHTML(bus.from)} to
${escapeHTML(bus.to)}</option>`
  ).join("");

  document.getElementById("bookingBus").innerHTML =
    '<option value="">Select a bus</option>' + options;

  document.getElementById("commentBus").innerHTML =
    '<option value="">Select a bus</option>' + options;
}
```

```
function selectBus(id) {
  selectedBusId = id;
  selectedSeats = [];
  document.getElementById("bookingBus").value = String(id);
  renderSeats();
  document.getElementById("booking").scrollIntoView({behavior: "smooth"});
}

document.getElementById("searchForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const from = document.getElementById("from").value.trim().toLowerCase();
  const to = document.getElementById("to").value.trim().toLowerCase();
  const date = document.getElementById("travelDate").value;

  if (date < new Date().toISOString().slice(0, 10)) {
    alert("Please choose today or a future date.");
    return;
  }

  const found = buses.filter(bus =>
    bus.from.toLowerCase() === from &&
    bus.to.toLowerCase() === to
  );

  document.getElementById("busList").innerHTML = found.length
    ? found.map(bus => `
      <div class="bus-card">
        <h3>${escapeHTML(bus.name)}</h3>
        <p>Category: ${escapeHTML(bus.category)}</p>
        <p>Route: ${escapeHTML(bus.from)} to ${escapeHTML(bus.to)}</p>
        <p>Departure: ${escapeHTML(bus.time)}</p>
        <p>Travel Date: ${escapeHTML(date)}</p>
        <p>Fare: ₹${bus.fare}</p>
```

```
            <p>Available seats: ${bus.totalSeats - bus.bookedSeats.length}</p>
            <button onclick="selectBus(${bus.id})">Select Bus</button>
          </div>
        `).join("")
        : "<p>No buses found for this route.</p>";
});

document.getElementById("bookingBus").addEventListener("change", function() {
  selectedBusId = Number(this.value) || null;
  selectedSeats = [];
  renderSeats();
});

function renderSeats() {
  const layout = document.getElementById("seatLayout");
  const bus = buses.find(b => b.id === selectedBusId);

  if (!bus) {
    layout.innerHTML = "<p>Please select a bus first.</p>";
    document.getElementById("selectedSeats").textContent = "Selected Seats: None";
    return;
  }

  layout.innerHTML = Array.from({length: bus.totalSeats}, (_, i) => {
    const seat = i + 1;
    const booked = bus.bookedSeats.includes(seat);
    const selected = selectedSeats.includes(seat);

    return `
      <button type="button"
        class="seat ${booked ? "booked" : ""} ${selected ? "selected" : ""}"
        ${booked ? "disabled" : ""}
        onclick="toggleSeat(${seat})">
        ${seat}
```

```
        </button>
      `;
    }).join("");

    document.getElementById("selectedSeats").textContent =
      "Selected Seats: " + (selectedSeats.join(", ") || "None");
}

function toggleSeat(seat) {
    if (selectedSeats.includes(seat)) {
      selectedSeats = selectedSeats.filter(s => s !== seat);
    } else {
      selectedSeats.push(seat);
    }

    renderSeats();
}

document.getElementById("bookingForm").addEventListener("submit", function(e) {
    e.preventDefault();

    if (!currentUser) {
      alert("Please register and log in before booking.");
      return;
    }

    const busId = Number(document.getElementById("bookingBus").value);
    const bus = buses.find(b => b.id === busId);
    const passenger = document.getElementById("passengerName").value.trim();
    const email = document.getElementById("passengerEmail").value.trim();
    const age = Number(document.getElementById("passengerAge").value);
    const date = document.getElementById("travelDate").value;

    if (!bus || selectedSeats.length === 0) {
```

```
    alert("Select a bus and at least one seat.");
    return;
}

if (!passenger || !email || age < 1 || age > 120 || !date) {
    alert("Enter valid passenger details and travel date.");
    return;
}

if (selectedSeats.some(seat => bus.bookedSeats.includes(seat))) {
    alert("A selected seat is no longer available. Please select again.");
    renderSeats();
    return;
}

bus.bookedSeats.push(...selectedSeats);

const booking = {
    id: nextBookingId++,
    email: currentUser.email,
    passenger,
    bus: bus.name,
    busId: bus.id,
    date,
    seats: [...selectedSeats],
    fare: bus.fare * selectedSeats.length,
    status: "Confirmed"
};

bookings.push(booking);

showMessage("bookingMessage",
    "Demo booking confirmed! Booking ID: " + booking.id +
    ". Total fare: ₹" + booking.fare);
```

```
    selectedSeats = [];
    renderSeats();
    renderBuses();
    renderHistory();
    renderAdminBookings();
    this.reset();
});

function renderHistory() {
    const table = document.getElementById("historyTable");

    if (!currentUser) {
        table.innerHTML = '<tr><td colspan="8">Please log in to view bookings.</td></tr>';
        return;
    }

    const mine = bookings.filter(b => b.email === currentUser.email);

    table.innerHTML = mine.length ? mine.map(b => `
        <tr>
            <td>${b.id}</td>
            <td>${escapeHTML(b.passenger)}</td>
            <td>${escapeHTML(b.bus)}</td>
            <td>${escapeHTML(b.date)}</td>
            <td>${b.seats.join(", ")}</td>
            <td>₹${b.fare}</td>
            <td>${b.status}</td>
            <td>${b.status === "Confirmed"
                ? `<button onclick="cancelBooking(${b.id})">Cancel</button>`
                : "-"}</td>
        </tr>
    `).join("") : '<tr><td colspan="8">No bookings found.</td></tr>';
}
```

```
function cancelBooking(id) {
  const booking = bookings.find(b => b.id === id);

  if (!booking || booking.email !== currentUser.email) return;

  if (!confirm("Cancel this booking?")) return;

  booking.status = "Cancelled";

  const bus = buses.find(b => b.id === booking.busId);
  if (bus) {
    bus.bookedSeats = bus.bookedSeats.filter(
      seat => !booking.seats.includes(seat)
    );
  }

  renderHistory();
  renderSeats();
  renderBuses();
  renderAdminBookings();

  alert("Demo booking cancelled.");
}

document.getElementById("commentForm").addEventListener("submit", function(e) {
  e.preventDefault();

  if (!currentUser) {
    alert("Please log in before posting a comment.");
    return;
  }

  const busId = Number(document.getElementById("commentBus").value);
```

```
  const bus = buses.find(b => b.id === busId);
  const text = document.getElementById("commentText").value.trim();

  if (!bus || !text) {
    alert("Select a bus and enter a comment.");
    return;
  }

  comments.push({
    user: currentUser.name,
    bus: bus.name,
    text
  });

  document.getElementById("commentList").innerHTML = comments.map(c => `
    <div class="bus-card">
      <strong>${escapeHTML(c.user)}</strong> - ${escapeHTML(c.bus)}
      <p>${escapeHTML(c.text)}</p>
    </div>
  `).join("");

  this.reset();
});

document.getElementById("adminLogin").addEventListener("submit", function(e) {
  e.preventDefault();

  // Demonstration password only; not secure authentication.
  if (document.getElementById("adminPassword").value === "admin123") {
    document.getElementById("adminPanel").style.display = "block";
    renderAdminBookings();
    renderUsers();
    alert("Demo admin login successful.");
  } else {
```

```
    alert("Incorrect admin password.");
  }
});

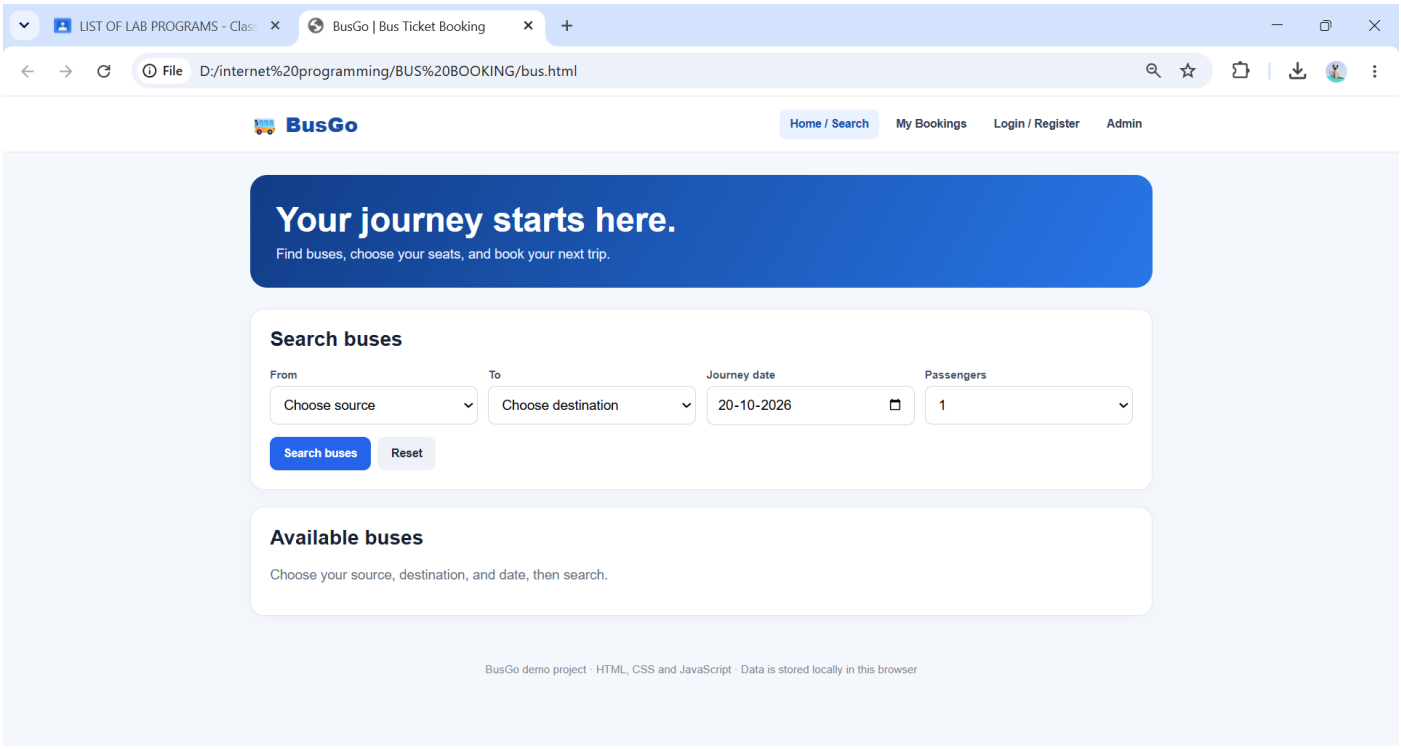
document.getElementById("busForm").addEventListener("submit", function(e) {
  e.preventDefault();

  buses.push({
    id: Math.max(0, ...buses.map(b => b.id)) + 1,
    name: document.getElementById("busName").value.trim(),
    category: document.getElementById("busCategory").value,
    from: document.getElementById("busFrom").value.trim(),
    to: document.getElementById("busTo").value.trim(),
    time: document.getElementById("busTime").value,
    fare: Number(document.getElementById("busFare").value),
    totalSeats: Number(document.getElementById("busSeats").value),
    bookedSeats: []
  });

  renderBuses();
  this.reset();
  alert("Bus added to this demo.");
});

function renderAdminBuses() {
  const area = document.getElementById("adminBusList");

  area.innerHTML = buses.map(bus => `
    <div class="bus-card">
      <h3>${escapeHTML(bus.name)}</h3>
      <p>${escapeHTML(bus.category)}: ${escapeHTML(bus.from)} to ${escapeHTML(bus.to)}</p>
      <p>Fare: ₹${bus.fare}</p>
      <p>Seats: ${bus.totalSeats}</p>
      <button onclick="editBus(${bus.id})">Edit Bus</button>
```


LIST OF LAB PROGRAMS - Clas
BusGo | Bus Ticket Booking
File
D:/internet%20programming/BUS%20BOOKING/bus.html
BusGo
Home / Search
My Bookings
Login / Register
Admin
Your journey starts here.
Find buses, choose your seats, and book your next trip.
Search buses
From
Choose source
To
Choose destination
Journey date
20-10-2026
Passengers
1
Search buses
Reset
Available buses
Choose your source, destination, and date, then search.
BusGo demo project · HTML, CSS and JavaScript · Data is stored locally in this browser